

3.2 Functions in Python

فنکشن (Function) ہدایات کا وہ بلاک ہے جو ایک بار لکھا جاتا ہے اور بار بار بلایا جا سکتا ہے۔

CS XII

Chapter 3: Programming Fundamentals

Lecture 09 of 17

★ Golden

3.2 Functions in Python ★ Golden

LEARN IT

When the same instructions are written again and again, that is **duplicate code**. Duplicate code makes a program longer, harder to read, and painful to update (you must fix every copy). A **function** lets you write the instructions **once** and run them whenever you call the function's name.

Benefit	Meaning
Code reusability	Write once, use anywhere.
Modularity	Break a big problem into small named tasks.
Readability	Shorter, cleaner files with meaningful names.
Maintainability	Fix a bug in one place and every call is fixed.

```
def greet():
    print("Hello, World!")

greet()
greet()
```

OUTPUT

```
Hello, World!
Hello, World!
```

The body is written once. Two calls run it twice. To change the message you edit one line inside `greet`.

Key components of a function

def → name → parameters → indented body → return (optional)

- **Keyword** — Python uses `def` to start a definition.
- **Function name** — the identifier you call, e.g. `greet`.
- **Parameters** — input names inside the parentheses. When you call the function, the values you pass are called **arguments**.
- **Body** — the indented block that runs on each call.
- **Return statement** — sends a result back to the caller (3.2.2).

```
def add(a, b):
    print(f"Adding {a} + {b}")
    return a + b

result = add(4, 6)
print(result)
```

OUTPUT

```
Adding 4 + 6
10
```

Here `a` and `b` are parameters. `4` and `6` are arguments. `return a + b` sends 10 back, and that 10 is stored in `result`.

WORKED EXAMPLE

Without a function you would copy the print-and-add lines every time you needed a sum. With a function, a new sum is one call: `add(10, 20)`, `add(1.5, 2.5)`.

REMEMBER IT

DRY — Don't Repeat Yourself. If you are about to copy-paste a block, wrap it in `def`. Name functions with verbs: `greet`, `add`, `save_grades`.

COMMON MISTAKE

Defining is not calling. After `def greet()`: nothing prints until you write `greet()`. Forgetting the parentheses — writing `greet` alone — does not run the body. The body must be indented; a line at column 0 is outside the function.

اردو میں سمجھیں

فنکشن (Function) ہدایات کا وہ بلاک ہے جو ایک بار لکھا جاتا ہے اور بار بار بلایا جا سکتا ہے۔ اس سے کوڈ مختصر، ماڈیولر اور آسان اصلاح والا بنتا ہے۔ تعریف `def` سے شروع ہوتی ہے، نام اور پیرامیٹرز قوسین میں ہوتے ہیں، جسم انڈینٹ ہوتا ہے، اور `return` نتیجہ واپس بھیج سکتا ہے۔ تعریف کرنے سے فنکشن چلتا نہیں جب تک آپ اسے کال نہ کریں۔

EXAM TIP

Write the four benefits (reusability, modularity, readability, maintainability) as separate bullets — that is a standard short question. For long questions, define a function, label `def`, name, parameters, body and `return`, then show one call. "What is duplicate code and how do functions solve it?" is also common.

CHECK YOURSELF

1. What keyword starts a function definition?
2. How many times does `Hello, World!` print in the `greet` program above, and why?

Answers: 1. `def`. 2. Twice, because `greet()` is called twice.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A function is:

- a. A named reusable block
- b. A random line in main only
- c. A pie
- d. A cable

MCQ 2. You define with:

- a. `def name(params):`
- b. `function name {}`
- c. `fun name`
- d. `lambda` only always

MCQ 3. A call is:

- a. `name(arguments)`
- b. `def` again
- c. `import name` always
- d. `print` the source only

MCQ 4. Duplicate code is bad because:

- a. Fixes must be repeated; more bugs
- b. Python forbids functions
- c. It is faster always
- d. It is required by BIEK

MCQ 5. This is ★: expect:

- a. Write a small function + call + output
- b. Only define HCI
- c. Only OSI
- d. Only ER

MCQ 6. Parameters are:

- a. Placeholders in the definition
- b. The values at the call (arguments)
- c. Always globals
- d. Always files

Extra short questions

1. Define function, parameter, argument.
2. Why functions (three reasons).
3. Show a 3-line add function.
4. What happens if you `def` but never call?

Extra long question

1. Write functions to compute area of rectangle and circle, call both from main, show output. Explain `def` vs `call`. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Reusable block; names in `def`; values you pass.
2. Reuse, clarity, testing, less duplication.
3. `def add(a,b): return a+b`
4. Nothing runs — no output.

Long

1. Two `defs`, two calls, printed numbers. Mention `return` vs `print` inside. $\pi=3.14$ or `math.pi`.